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Semiconductor device with two-phase cooling structure

Abstract

A semiconductor device includes a semiconductor chip including a semiconductor integrated circuit; a cooling channel including a surface of the semiconductor chip and configured to provide a passage for a coolant to cool the semiconductor chip; and a plurality of flexible capillary patterns on the surface of the semiconductor chip inside the cooling channel and configured to move the coolant by capillary action, wherein each capillary pattern of the plurality of flexible capillary patterns may include a first portion in a length direction of the capillary pattern that contacts and is supported by the surface of the semiconductor chip, and a second portion in the length direction that is spaced apart from and unsupported by the surface of the semiconductor chip, and a curvature of the second portion of each capillary pattern of the plurality of flexible capillary patterns changes according to temperature.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2022-0179735, filed on Dec. 20, 2022, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(2) The disclosure relates to a semiconductor device with a cooling structure.

2. Description of Related Art

(3) In order to remove heat generated from electronic devices, an air-cooling device has been mainly used. As the power density of electronic devices gradually increases, the use of cooling devices is increasing to cope with the increased heat generation. Moreover, in the case of a data center, in order to reduce the amount of power used, interest in a next-generation cooling method with high efficiency, such as a liquid cooling device, is gradually increasing. Depending on a temperature range of a part where heat is generated, the cooling method includes a single-phase liquid cooling method without a phase change of a coolant and a two-phase liquid cooling method involving a phase change of a coolant. The two-phase cooling method has a higher calorific value range than the single-phase cooling method.

SUMMARY

(4) Provided is a semiconductor device employing a two-phase liquid cooling method.

(5) Provided is a semiconductor device employing a two-phase liquid cooling method capable of effectively supplying a liquid coolant to a heat transfer surface.

(6) Provided is a semiconductor device employing a two-phase liquid cooling method capable of coping with a decrease in cooling efficiency due to vapor adsorption to a heat transfer surface.

(7) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of embodiments of the disclosure.

(8) According to an aspect of an example embodiment, a semiconductor device includes: a semiconductor chip including a semiconductor integrated circuit; a cooling channel including a surface of the semiconductor chip and configured to provide a passage for a coolant to cool the semiconductor chip; and a plurality of flexible capillary patterns on the surface of the semiconductor chip inside the cooling channel and configured to move the coolant by capillary action, wherein each capillary pattern of the plurality of flexible capillary patterns may include a first portion in a length direction of the capillary pattern that contacts and is supported by the

surface of the semiconductor chip, and a second portion in the length direction that is spaced apart from and unsupported by the surface of the semiconductor chip, and a curvature of the second portion of each capillary pattern of the plurality of flexible capillary patterns changes according to temperature.

(9) Each capillary pattern of the plurality of flexible capillary patterns further may include a first layer having a first thermal expansion coefficient and a second layer having a second thermal expansion coefficient that is different from the first thermal expansion coefficient.

(10) The second portion of each capillary pattern of the plurality of flexible capillary patterns may have a roll shape.

(11) The surface of the semiconductor chip on which the plurality of flexible capillary patterns are formed may be an upper surface of the semiconductor chip.

(12) Each capillary pattern of the plurality of flexible capillary patterns may include a first layer having a first thermal expansion coefficient and a second layer having a second thermal expansion coefficient that is less than the first thermal expansion coefficient, and the first layer may be adjacent to the surface of the semiconductor chip, and the second layer may be disposed on the first layer.

(13) Each capillary pattern of the plurality of flexible capillary patterns may include a plurality of curved portions that are convex in a direction opposite to the surface of the semiconductor chip.

(14) Each curved portion of the plurality of curved portions extends in a width direction of the capillary pattern that is perpendicular to the length direction.

(15) Each capillary pattern of the plurality of flexible capillary patterns may include a plurality of protrusions configured to prevent adhesion with the surface of the semiconductor chip.

(16) The plurality of protrusions may be on a surface of the capillary pattern facing the surface of the semiconductor chip.

(17) Each capillary pattern of the plurality of flexible capillary patterns may include a surface roughness structure on a surface of the capillary pattern facing to the surface of the semiconductor chip.

(18) The cooling channel may include an immersed unit immersed from an upper surface of the semiconductor chip, the upper surface of the semiconductor chip may include a bottom of the immersed unit, and the capillary pattern may be on the bottom of the immersed unit.

(19) The semiconductor device may further include a housing surrounding the semiconductor chip, and the cooling channel may be in a space between the semiconductor chip and the housing.

(20) According to an aspect of an example embodiment, a semiconductor device includes: a semiconductor chip including a semiconductor integrated circuit; a cooling channel including a surface of the semiconductor chip and configured to provide a passage for a coolant to cool the semiconductor chip; and a flexible capillary pattern formed on the surface of the semiconductor chip to move the coolant by capillary action, wherein at least a portion of the flexible capillary pattern has a roll shape, and the flexible capillary pattern may include a first layer having a first thermal expansion coefficient and a second layer having a second thermal expansion coefficient that different from the first thermal expansion coefficient.

(21) A curvature of the flexible capillary pattern may change with temperature.

(22) The flexible capillary pattern may include an anti-distortion unit configured to prevent distortion of the flexible capillary pattern in a width direction of the flexible capillary pattern.

(23) The anti-distortion unit may include a plurality of curved portions that are convex in a direction opposite to the surface of the semiconductor chip, and extend in a width direction of the flexible capillary pattern.

(24) The semiconductor may further include a plurality of protrusions formed on a surface of the flexible capillary pattern facing the surface of the semiconductor chip.

(25) The flexible capillary pattern may include a surface roughness structure formed on a surface of the flexible capillary pattern facing the surface of the semiconductor chip.

- (26) The cooling channel may include an immersed unit immersed from an upper surface of the semiconductor chip, the upper surface of the semiconductor chip may include a bottom of the immersed unit, and the flexible capillary pattern may be on the bottom of the immersed unit.
- (27) The semiconductor device may further include a housing surrounding the semiconductor chip, and the cooling channel may be in a space between the semiconductor chip and the housing.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a schematic diagram of a configuration of a semiconductor device according to an embodiment;
- (3) FIGS. 2A to 2C are plan views showing various examples of arrangement of a plurality of capillary patterns;
- (4) FIG. 3 is a cross-sectional view of a capillary pattern in a length direction, according to an embodiment;
- (5) FIGS. 4A to 4E show example shapes of capillary patterns;
- (6) FIG. 5A shows a capillary pattern according to an embodiment;
- (7) FIG. 5B shows a capillary pattern according to an embodiment;
- (8) FIG. 5C shows a capillary pattern according to an embodiment; and
- (9) FIG. 6 is a schematic diagram of a configuration of a semiconductor device according to an embodiment.

DETAILED DESCRIPTION

(10) Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, embodiments are merely described below, by referring to the figures, to explain aspects. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

(11) Hereinafter, reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings. In the drawings, like reference numerals denote like elements and sizes of constituent elements may be exaggerated for convenience of explanation and clarity. Embodiments of the disclosure are capable of various modifications and may be embodied in many different forms. When a position of an element is described using an expression “above” or “on”, the position of the element may include not only the element being “immediately on/under/left/right in a contact manner” but also being “on/under/left/right in a non-contact manner”. Singular forms are intended to include the plural forms as well unless the context clearly indicates otherwise. When a part “comprises” or “includes” an element in the specification, unless otherwise defined, it is not excluding other elements but may further include other elements. In the specification, the term “above” and similar directional terms may be applied to both singular and plural. With respect to operations that constitute a method, the operations may be performed in any appropriate sequence unless the sequence of operations is clearly described or unless the context clearly indicates otherwise. The operations may not necessarily be performed in the order of sequence. Also, in the specification, the term “units” or “. . . modules” denote units or modules that process at least one function or operation, and may be realized by hardware, software, or a

combination of hardware and software. Connections or connection members of lines between components shown in the drawings illustrate functional connections and/or physical or circuit connections, and the connections or connection members can be represented by replaceable or additional various functional connections, physical connections, or circuit connections in an actual apparatus. The use of any and all examples, or exemplary language provided herein, is intended merely to better illuminate the disclosure and does not pose a limitation on the scope of the disclosure unless otherwise claimed.

(12) An efficient cooling system is required to solve a cooling problem that has been a limiting factor in the performance of electronic devices including semiconductor devices. In the case of high performance computing (HPC) fields and semiconductor devices to which stacked three-dimensional semiconductor integrated circuits are applied, a cooling system capable of responding to an increase in power density and an increase in heat generation due to high integration is required. In order to meet these demands, a two-phase liquid cooling system capable of utilizing latent heat of vaporization of a coolant may be applied to an integrated circuit device. The two-phase liquid cooling method may include immersion cooling, spray cooling, and jet impingement cooling.

(13) Immersion cooling is a method of cooling an electronic device by immersing the electronic device in a bath containing a liquid coolant. The immersion cooling method is the most widely used method, but because the entire electronic device needs to be immersed in a liquid coolant, the usable coolant is limited to a dielectric coolant. Therefore, there are problems in that this method requires high management costs and is not environmentally friendly. In addition, when a film boiling phenomenon occurs in which vapor of the coolant is adsorbed to a heating surface, the vapor prevents heat transfer from the heating surface to the coolant, thereby reducing cooling efficiency. In a boiling graph, this phenomenon is called critical heat flux (CHF). It is necessary to improve the CHF to cool a larger heat flux.

(14) Spray cooling is a method of atomizing a liquid coolant and spraying the coolant on a heating surface and has high cooling efficiency compared to the amount of liquid coolant used. However, a spray cooling device requires a pumping device capable of operating at high pressure and requires continuous maintenance of spray nozzles. The spray cooling also has a problem of lowering cooling efficiency due to a film boiling phenomenon.

(15) Jet impingement cooling is a cooling method in which a liquid coolant is injected to a heating surface at a high speed. It shows high cooling efficiency like the spray cooling method but has a disadvantage of requiring several injectors to evenly cool the heating surface, and in addition, even if several injectors are used, there may be a blind area where the jet of liquid coolant does not reach. In addition, the jet impingement cooling also has a problem of a decrease in cooling efficiency due to a film boiling phenomenon.

(16) The two-phase liquid cooling system according to an embodiment supplies a liquid coolant (refrigerant) to a semiconductor chip by using a capillary action to cool an object to be cooled, for example, a semiconductor chip. For this purpose, a capillary pattern is disposed in a cooling channel. The capillary pattern is formed on a surface of a semiconductor chip included in the cooling channel. The method of supplying a liquid coolant using a capillary phenomenon uses a small amount of liquid coolant and, in some cases, may supply a liquid coolant without a driving device, such as a pump, so the cooling structure may be miniaturized and is suitable for semiconductor devices including semiconductor chips. In a cooling channel, the liquid coolant absorbs heat from a heat transfer surface adjacent to a heat source and vaporizes into vapor. When vapor is stagnant or attached near the heat transfer surface and a vapor film is generated, cooling efficiency may be reduced and dry-out may occur locally. According to a two-phase cooling system of the disclosure, a capillary pattern, the curvature of which changes with temperature, is employed. The capillary pattern may include a plurality of layers having different thermal expansion coefficients. The capillary pattern may have a cantilever shape. As temperature changes,

a bending direction and curvature of the capillary pattern change. Accordingly, the supply speed and amount of a liquid coolant may be adjusted according to the temperature. In addition, the cantilever-shaped capillary pattern is vibrated by vibration energy generated when the liquid coolant is vaporized. Vapors near the capillary pattern may be dislodged from the heat transfer surface by vibrational energy. Therefore, the liquid coolant may be stably supplied to the vicinity of the heat source, and the stagnation and adhesion of vapor near the heat transfer surface may be reduced or prevented.

(17) In the following description (see, e.g., FIG. 1), a first direction (X) denotes one of the directions parallel to an upper surface of the semiconductor chip. A second direction (Z) denotes a thickness direction of the semiconductor chip. A third direction (Y) denotes a direction orthogonal to the first direction (X) among directions parallel to the upper surface of the semiconductor chip. FIG. 1 is a schematic diagram of a configuration of a semiconductor device 1 according to an embodiment. Referring to FIG. 1, the semiconductor device (integrated circuit device) 1 according to an embodiment may include at least one semiconductor chip (integrated circuit die) 100, a cooling channel 200, and a capillary pattern 300.

(18) The number of semiconductor chips may be one or more. As an example, the at least one semiconductor chip may include the semiconductor chip 100. The semiconductor chip 100 may include a substrate 110 and a semiconductor integrated circuit 120 formed on one surface of the substrate 110. The semiconductor chip 100 may include various semiconductor chips. For example, the semiconductor chip 100 may include a memory chip (die) including a memory integrated circuit, a logic chip (die) including a logic integrated circuit, a central processing unit (CPU) chip, a graphics processing unit (GPU) chip (die), an application specific integrated circuit (ASIC) chip (die), and the like. In order to implement the semiconductor device 1 of a small form factor, the semiconductor chip 100 may be a wafer-level semiconductor chip. In this case, the substrate 110 may be a wafer.

(19) As an example, the at least one semiconductor chip may further include semiconductor chips 100a and 100b. The semiconductor chips 100a and 100b may be arranged in a transverse direction with respect to the semiconductor chip 100, for example, in a first direction X. Each of the semiconductor chips 100a and 100b may include a substrate and a semiconductor integrated circuit formed on one surface of the substrate. The semiconductor chips 100a and 100b may include a memory chip (die) including a memory integrated circuit, a logic chip (die) including a logic integrated circuit, a CPU chip (die), a GPU chip (die), an ASIC chip (die), or the like. In order to implement the semiconductor device 1 of a small form factor, the semiconductor chips 100a and 100b may be wafer-level semiconductor chips. In this case, a substrate of each of the semiconductor chips 100a and 100b may be a wafer. For example, the semiconductor chip 100 may be a CPU or GPU. The semiconductor chips 100a and 100b may be memory chips (die), for example, high bandwidth memory (HBM) chips (die).

(20) The semiconductor chips 100, 100a, and 100b are mounted on a sub-substrate 1000S, and an assembly of the semiconductor chips 100, 100a, and 100b and the sub-substrate 1000S may be mounted, for example, on a printed circuit board 1000 by solder balls. Lower surfaces of the semiconductor chips 100, 100a, and 100b and an upper surface of the sub-substrate 1000S may be electrically passivated by, for example, a passivation member 180. A lower surface of the sub-substrate 1000S and an upper surface of the printed circuit board 1000 may be electrically passivated by a passivation layer 190. The passivation member 180 and the passivation layer 190 may function as a sealing material for a coolant.

(21) A two-phase liquid cooling structure may include the cooling channel 200. The cooling channel 200 forms a passage for a coolant (refrigerant) to cool the at least one semiconductor chip 100. The cooling channel 200 may include at least a surface of the semiconductor chip 100, for example, an upper surface 101. In an embodiment, the cooling channel 200 includes upper surfaces 101a, 101, and 101b of the semiconductor chips 100a, 100, and 100b to cool the semiconductor

chips **100a**, **100**, and **100b**. For example, the semiconductor chips **100**, **100a**, and **100b** may be packaged by a package housing **700**. The package housing **700** surrounds the semiconductor chips **100**, **100a**, and **100b**. The cooling channel **200** may be formed by a space between the package housing **700** and the semiconductor chips **100**, **100a**, and **100b**. At least one of the upper surfaces **101a**, **101**, and **101b** of the semiconductor chips **100a**, **100**, and **100b** is in contact with a coolant accommodated in the cooling channel **200**. One or more openings communicating with the cooling channel **200**, for example, openings **701**, **702**, and **703** may be provided in an upper wall **710** of the package housing **700**. The cooling channel **200** is connected to a condenser (or radiator) **800** by the openings **701**, **702**, and **703**. The cooling channel **200** and the condenser (or radiator) **800** form a two-phase liquid cooling system. A coolant storage unit may be provided between the condenser **800** and the opening **702**. The liquid coolant condensed in the condenser **800** is received in the coolant storage unit and may be supplied from the coolant storage unit to the cooling channel **200** through the opening **702**. The upper wall **710** of the package housing **700** forming an upper wall of the cooling channel **200** may be a cold plate having a heat dissipation function. The cold plate may function as a condenser (or radiator) **800**. In this case, the openings **701**, **702**, and **703** may not be provided, and the coolant in the cooling channel **200** dissipates heat to the outside by heat exchange with the upper wall **710** of the package housing **700**. A cold plate, such as a heat sink, may be provided outside the upper wall **710** of the package housing **700**.

(22) The capillary pattern **300** is provided inside the cooling channel **200**. The capillary pattern **300** may be provided on a heat transfer surface. The capillary pattern **300** moves a liquid coolant along the heat transfer surface by capillary action. That is, the capillary pattern **300** moves the liquid coolant inside the cooling channel **200** to the heat transfer surface. In addition, because the capillary pattern **300** increases a surface area of the heat transfer surface, heat exchange efficiency may be increased.

(23) Because the liquid coolant inside the cooling channel **200** contacts the surface of the semiconductor chip **100** and receives heat from the semiconductor chip **100**, the surface of the semiconductor chip **100** performs as a heat transfer surface. The capillary pattern **300** may be provided on the surface of the semiconductor chip **100**, for example, the upper surface **101**. The liquid coolant may be effectively supplied to the surface of the semiconductor chip **100**, for example, the upper surface **101** by the capillary phenomenon, and the surface area of the heat transfer surface may be increased. In the embodiment, the capillary pattern **300** is provided on the upper surfaces **101**, **101a**, and **101b** of the semiconductor chips **100**, **100a**, and **100b**. Of course, the capillary pattern **300** may be provided on the upper surface **101** of the semiconductor chip **100** and may not be provided on the upper surfaces **101a** and **101b** of the semiconductor chips **100a** and **100b**.

(24) A plurality of capillary patterns **300** may be arranged on the surfaces, for example, the upper surfaces **101**, **101a**, and **101b** of the semiconductor chips **100**, **100a**, and **100b**. Hereinafter, as an example, a form in which a plurality of capillary patterns **300** are arranged on the surface of the semiconductor chip **100**, for example, the upper surface **101**, are described, and the arrangement of the plurality of capillary patterns **300** may be the same on the surfaces, for example, the upper surfaces **101a** and **101b** of the semiconductor chips **100a** and **100b**. FIGS. 2A to 2C are plan views showing various examples of the arrangement of a plurality of capillary patterns **300**.

(25) First, referring to FIG. 2A which shows at least some of the capillary patterns **300**, each capillary pattern **300** has a length in a first direction (X), a thickness in a second direction (Z), and a width in a third direction (Y). The plurality of capillary patterns **300** are two-dimensionally arranged on the upper surface **101** of the semiconductor chip **100**. A capillary channel **210a** is formed between two capillary patterns **300** adjacent to each other in the first direction (X). A capillary channel **210b** extending in the first direction (X) is formed between two capillary patterns **300** adjacent to each other in the third direction (Y). The plurality of capillary channels **210b** communicate with each other through the plurality of capillary channels **210a**. A liquid coolant

may be moved by a capillary force through the capillary channel **210a** and the capillary channel **210b**.

(26) Referring to FIG. 2B which shows at least some of the capillary patterns **300**, each capillary pattern **300** has a length in the third direction (Y), a thickness in the second direction (Z), and a width in the first direction (X). The plurality of capillary patterns **300** are two-dimensionally arranged on the upper surface **101** of the semiconductor chip **100**. A capillary channel **210c** extending in the third direction (Y) is formed between two capillary patterns **300** adjacent to each other in the first direction (X). A capillary channel **210d** is formed between two capillary patterns **300** adjacent to each other in the third direction (Y). The plurality of capillary channels **210c** communicate with each other through the plurality of capillary channels **210d**. The liquid coolant may be moved by capillary force through the capillary channel **210c** and the capillary channel **210d**.

(27) Referring to FIG. 2C, the arrangement shown in FIG. 2A and the arrangement shown in FIG. 2B are combined. As a result, the capillary channel **210b** extending in the first direction (X), the capillary channel **210c** extending in the third direction (Y), and the capillary channel **210a** and the capillary channel **210d** that communicate with the capillary channel **210b** and the capillary channel **210c** are formed. The liquid coolant may be moved by capillary force through the capillary channels **210a**, **210b**, **210c**, and **210d**.

(28) The arrangement of the capillary pattern **300** is not limited to the examples shown in FIGS. 2A to 2C. The capillary pattern **300** may extend in an arbitrary direction within the transverse plane, and the plurality of capillary patterns **300** may be arranged in any direction as long as they form capillary channels generating capillary forces. Lengths of the plurality of capillary patterns **300** may all be the same, and some of the capillary patterns **300** may have different lengths from the rest of the capillary patterns **300**.

(29) FIG. 3 is a cross-sectional view of a capillary pattern in a length direction, according to an embodiment. Referring to FIG. 3, the capillary pattern **300** has a cantilever shape. One end (first portion) **301** of the capillary pattern **300** in the length direction is fixed to a surface of the semiconductor chip **100**, for example, the upper surface **101**, and the other end (second portion) **302** is free. The capillary pattern **300** may have a flexible cantilever shape, a curvature of which changes according to temperature. A distance between the capillary pattern **300** and the surface of the semiconductor chip **100**, for example, the upper surface **101** changes according to temperature. This means that the shape and size of the capillary channel formed between the capillary pattern **300** and the upper surface **101** of the semiconductor chip **100** changes according to temperature. Of course, as described below with reference to FIGS. 4D and 4E, in the case when the capillary pattern **300** is in a roll form, the shape and size of the capillary channel inside the roll shape of the capillary pattern **300** also change according to temperature. Accordingly, according to temperature change, an appropriate amount of the liquid coolant LC may be effectively supplied to the upper surface **101** of the semiconductor chip **100**, which is a heat transfer surface.

(30) The capillary pattern **300** may include a plurality of layers stacked in a thickness direction, that is, in the second direction (Z). At least one of the plurality of layers has a different thermal expansion coefficient than the other layers. For example, the capillary pattern **300** may include a first layer **310** and a second layer **320**. The first layer **310** is close to the surface of the semiconductor chip **100**, for example, the upper surface **101**, and the second layer **320** is far from the surface of the semiconductor chip **100**, for example, the upper surface **101**. In other words, a distance from the first layer **310** to the upper surface **101** is smaller than a distance from the second layer **320** to the upper surface **101**. The second layer **320** may be formed by being stacked on the first layer **310**. The first layer **310** may be spaced apart from the upper surface **101** of the semiconductor chip **100**. Accordingly, a gap **330** is formed between the surface of the semiconductor chip **100**, for example, the upper surface **101** and the capillary pattern **300**. This gap may function as a capillary channel creating a capillary force. Accordingly, a coolant may be more

easily supplied to the surface of the semiconductor chip **100**, for example, the upper surface **101**. Also, because a surface area of the surface of the semiconductor chip **100**, for example, the upper surface **101** directly exposed to the coolant is increased, the cooling efficiency may be improved.

(31) The first layer **310** and the second layer **320** include materials having different thermal expansion coefficients from each other. The thermal expansion coefficient of the first layer **310** may be greater or less than that of the second layer **320**. Because the amount of expansion of the first layer **310** and the second layer **320** is different due to the difference in thermal expansion coefficient between the first layer **310** and the second layer **320**, the cantilever-shaped capillary pattern **300** is bent. The curvature of the capillary pattern **300** varies according to temperature. The first layer **310** and the second layer **320** may include, for example, materials having different thermal expansion coefficients, such as metals, oxides, nitrides, etc. The metal may include, for example, Ti, Ta, Cu, Cr, Al, Mo, Ni, W, and the like. The oxides may include, for example, SiO₂, Al₂O₃, and the like. The nitride may include, for example, SiN, AlN, TiN, TaN, and the like. In the case of a combination in which the thermal expansion coefficient of the first layer **310** is greater than that of the second layer **320**, the first layer **310** may include a metal having a greater thermal expansion coefficient, and the second layer **320** may include a metal, oxide, nitride, etc. having a relatively small thermal expansion coefficient. Examples of combination of the first layer **310**/second layer **320** may include Al/Mo, Cu/Ni, Al/SiN, Al/SiO₂, etc. Conversely, examples of combination of the first layer **310**/second layer **320** when the thermal expansion coefficient of the second layer **320** is greater than that of the first layer **310** include Mo/Al, Cr/Al, TiN/Al, SiN/Cu, SiO₂/Mo, etc.

(32) The capillary pattern **300** of this type may be formed by using various manufacturing methods. As an example, a sacrificial layer is formed on the surface of the semiconductor chip **100**, for example, on the upper surface **101**. The first layer **310** and the second layer **320** are stacked on the sacrificial layer. Then, the first layer **310** and the second layer **320** are patterned according to a shape of the capillary pattern **300**. Then, when the sacrificial layer is removed, the capillary pattern **300** according to an embodiment may be formed. The method of forming the capillary pattern **300** is not limited to the method described above.

(33) Depending on the combination of thermal expansion coefficients of the first layer **310** and the second layer **320**, a bending direction and amount of bending of the capillary pattern **300** according to temperature may be determined. A combination of thermal expansion coefficients of the first layer **310** and the second layer **320** and parameters, such as a thickness, length, and width of the capillary pattern **300** may be appropriately determined considering a calorific value of the semiconductor chip **100** and a coolant flow rate required to cool the semiconductor chip **100**.

(34) Hereinafter, a behavior of the capillary pattern **300** according to a difference in thermal expansion coefficient between the first layer **310** and the second layer **320** is described. In the following description, as an example, a case when the capillary pattern **300** is formed on the upper surface **101** of the semiconductor chip **100** is described, and the behavior of the capillary patterns **300** installed on the upper surfaces **101a** and **101b** of the semiconductor chips **100a** and **100b** may also be the same.

(35) FIGS. 4A to 4E show example shapes of the capillary pattern **300**. In FIGS. 4A to 4E, the capillary pattern **300** is schematically shown in a cantilever shape in which one end **301** is supported on the upper surface **101** of the semiconductor chip **100** and the other end **302** is free. The capillary pattern **300** may have, for example, any one of the shapes shown in FIGS. 4A to 4E. An initial shape of the capillary pattern **300** may be determined by a difference in residual stress and stiffness between the first layer **310** and the second layer **320**, and the thickness, width, length, and initial temperature of the capillary pattern **300**.

(36) As an example, the capillary pattern **300** may initially be substantially parallel to the upper surface **101** of the semiconductor chip **100** as shown in FIG. 4A. As an example, in the capillary pattern **300**, the thermal expansion coefficient of the first layer **310** is greater than that of the

second layer **320**. In this state, a space **330a** is formed between the capillary pattern **300** and the upper surface **101** of the semiconductor chip **100**, and the space **330a** functions as a capillary channel. The space **330a** is filled with a liquid coolant LC introduced by the capillary phenomenon. In addition, a space between the adjacent capillary patterns **300** may also act as a capillary channel and may be filled with the liquid coolant LC. As temperature rises, the other end **302** of the capillary pattern **330** is separated from the upper surface **101** of the semiconductor chip **100** and is bent as shown by a solid line in FIG. **4B**. Accordingly, a space **330b** is created between the capillary pattern **300** and the upper surface **101** of the semiconductor chip **100**. A size of the space **330b** is greater than the space **330a** shown in FIG. **4A**. The space **330b** serves as a capillary channel, and an inside of the space **330b** is filled with the liquid coolant LC. In addition, a space between adjacent capillary patterns **300** may also act as a capillary channel and may be filled with the liquid coolant LC. As the temperature further rises, the bending of the capillary pattern **300** increases as shown in FIG. **4C**, and thus, a larger space **330c** serving as a capillary channel may be formed between the capillary pattern **300** and the upper surface **101** of the semiconductor chip **100**. In addition, a space between adjacent capillary patterns **300** may also act as a capillary channel and may be filled with the liquid coolant LC. A curvature of the capillary pattern **300** shown in FIG. **4C** is greater than the curvature of the capillary pattern **300** shown in FIG. **4B**. As the temperature increases, the curvature of the capillary pattern **300** may gradually increase. The capillary pattern **300** may be in the form of a roll as shown in FIG. **4D**. At this time, at least one of the space **330** between the roll-shaped capillary pattern **300** and the upper surface **101** of the semiconductor chip **100** and an inner space **330e** of the roll-shaped capillary pattern **300** may act as a capillary channel, and the liquid coolant LC may be filled in the space **330d** and/or the inner space **330e** serving as a capillary channel. In addition, the space between adjacent capillary patterns **300** may also act as a capillary channel and may be filled with the liquid coolant LC. As shown in FIG. **4E**, the capillary pattern **300** may be in the form of a roll wound one or more times. At this time, at least one of a space **330f** between the roll-shaped capillary pattern **300** and the upper surface **101** of the semiconductor chip **100**, a space **330g** between rolls of the roll-shaped capillary pattern **300**, and an innermost space **330h** of the roll-shaped capillary pattern **300** may act as capillary channels, and among the spaces **330f**, **330g**, and **330h**, a space that acts as a capillary channel may be filled with the liquid coolant LC. In addition, the space between adjacent capillary patterns **300** may also act as a capillary channel and may be filled with the liquid coolant LC. The capillary pattern **300** is changed to one of the shapes shown in FIGS. **4A** to **4E** according to the temperature of the liquid coolant LC, and accordingly, the gap between the surface of the semiconductor chip **100**, for example, the upper surface **101** and the capillary pattern **300** is changed, and thus, capillary force may be generated to continuously supply the liquid coolant LC to the upper surface **101** of the semiconductor chip **100**.

(37) When the thermal expansion coefficient of the first layer **310** is greater than that of the second layer **320**, as the temperature rises, as described above, the other end **302**, which is a free end, of the capillary pattern **300** rotates in a counterclockwise direction, and thus, the curvature of the capillary pattern **300** gradually increases. When the thermal expansion coefficient of the first layer **310** is less than that of the second layer **320**, as the temperature rises, the other end **302**, which is a free end, of the capillary pattern **300** rotates in a clockwise direction, and thus, the curvature of the capillary pattern **300** gradually increases.

(38) The capillary pattern **300** may have a bent shape initially, for example, as any one of the shapes shown in FIGS. **4B** to **4E** due to the combination of thermal expansion coefficients of the first layer **310** and the second layer **320** and residual stress during manufacture. Of course, the initial shape of the capillary pattern **300** may be a shape bent in an opposite direction to the shape shown in FIGS. **4B** to **4E**.

(39) For example, the initial shape of the capillary pattern **300** may be a roll shape shown in FIG. **4C**. When the thermal expansion coefficient of the first layer **310** is greater than that of the second

layer **320**, the capillary pattern **300** may be bent in the form shown in FIGS. **4D** and **4E** as a temperature rises. When the coefficient of thermal expansion of the first layer **310** is less than the coefficient of thermal expansion of the second layer **320**, the capillary pattern **300** may unfold in the form shown in FIGS. **4B** and **4A** as the temperature rises.

(40) For example, the initial shape of the capillary pattern **300** may be a roll shape shown in FIG. **4D**. When the thermal expansion coefficient of the first layer **310** is greater than that of the second layer **320**, the capillary pattern **300** may be bent as shown in FIG. **4E** as the temperature increases. When the thermal expansion coefficient of the first layer **310** is less than that of the second layer **320**, the capillary pattern **300** may unfold in the form shown in FIGS. **4C**, **4B**, and **4A** as the temperature rises. Of course, the initial shape of the capillary pattern **300** may be the roll shape shown in FIG. **4E**.

(41) An initial shape of the capillary pattern **300**, a combination of residual stress and stiffness of the first layer **310** and the second layer **320**, a combination of thermal expansion coefficients, parameters such as thickness, width, and length of the capillary pattern **300** may be determined to be able to supply a liquid coolant LC to the upper surface **101** of the semiconductor chip **100** at a flow rate sufficient enough to cool the semiconductor chip **100** through a capillary channel created by bending of the capillary pattern **300** considering the calorific value of the semiconductor chip **100**.

(42) A liquid coolant LC is filled inside capillary channels formed by the capillary pattern **300**. Because the capillary pattern **300** is formed on a surface of the semiconductor chip **100**, for example, the upper surface **101** which is a heat exchange surface, the liquid coolant LC may be continuously supplied to the heat exchange surface by a capillary phenomenon. The liquid coolant LC absorbs heat from a heat source inside the semiconductor chip **100**, for example, the semiconductor integrated circuit **120**. The heated liquid coolant LC is cooled by heat exchange with the outside in a condenser (or radiator) **800** or a cold plate and is re-supplied to the cooling channel **200**. The liquid coolant LC in the capillary channel may vaporize into a coolant vapor VC. The coolant vapor VC moves along the cooling channel **200** to the condenser (or radiator) **800**. The coolant vapor VC is phase-changed into a liquid coolant LC in the condenser (or radiator) **800**, and then re-supplied to the cooling channel **200**. Of course, the coolant vapor VC may be phase-changed into the liquid coolant LC by heat exchange with the outside in the cold plate, and then supplied to the cooling channel **200** again. A space inside the capillary channel from which the coolant vapor VC leaves is filled with the liquid coolant LC supplied by capillary force. The semiconductor chips **100**, **100a**, and **100b** may be effectively cooled by using latent heat of evaporation during a phase change process of the liquid coolant.

(43) Local overheating of the semiconductor chip **100** may cause local dry-out on the inside of the cooling channel **200**, for example, on the upper surface **101** of the semiconductor chip **100**, which is a heat transfer surface. Because the liquid coolant LC is not supplied to the area where dry-out occurs, the area where the dry-out occurs may be further overheated. A curvature of the cantilever-shaped capillary pattern **300** is changed according to the temperature of the liquid coolant LC, and thereby the size and shape of the capillary channel between the capillary pattern **300** and the upper surface **101** of the semiconductor chip **100** and/or the capillary channel of the inside of the capillary pattern **300** are also changed. By properly determining an initial shape of the capillary pattern **300**, a combination of thermal expansion coefficients of the first layer **310** and the second layer **320**, parameters such as thickness, width, and length of the capillary pattern **300**, an appropriate flow rate of the liquid coolant LC may be supplied to the upper surface **101** of the semiconductor chip **100**. In this way, the dry-out due to local overheating of the semiconductor chip **100** may be reduced or prevented.

(44) For effective cooling, coolant vapor VC generated inside the cooling channel **200** is needed to be discharged to the outside of the cooling channel **200**. The coolant vapor VC may be stagnant near a wall surface of the cooling channel **200**, for example, the upper surface **101** of the

semiconductor chip **100** adjacent to the semiconductor integrated circuit **120**, which is a heat source. When the coolant vapor VC near the upper surface **101** of the semiconductor chip **100** is bonded to each other and attached to the upper surface **101** of the semiconductor chip **100**, a vapor film may be formed. The stagnant vapor and vapor film may interfere with the contact between the liquid coolant LC and the upper surface **101** of the semiconductor chip **100**, resulting in an increase in thermal resistance, and as a result, a hot spot may be locally generated inside the semiconductor chip **100**. According to an embodiment, the cantilever-shaped capillary pattern **300** may cause coolant vapor VC to move in the capillary channel while the curvature changes according to temperature and may discharge the coolant vapor VC from the capillary channel. In addition, the movement of the capillary pattern **300** according to the change in curvature excites the liquid coolant LC, and the stagnant coolant vapor VC and/or the vapor film may be dispersed by the excited liquid coolant LC. In addition, the capillary pattern **300** is vibrated by the movement of the liquid coolant LC and the coolant vapor VC generated while boiling the liquid coolant LC, and the vibration of the capillary pattern **300** causes the movement of the coolant vapor VC around the capillary pattern **300**, and thus, the stagnant coolant vapor VC and/or vapor films may disperse. Accordingly, the liquid coolant LC may be stably supplied to the vicinity of the heat transfer surface, that is, to the upper surface **101** of the semiconductor chip **100**, and vapor stagnation and adhesion near the heat transfer surface may be reduced or prevented.

(45) FIG. 5A shows a capillary pattern **300** according to an embodiment. In order to clarify the characteristics of the embodiment, in FIG. 5A, the capillary pattern **300** is simply shown without distinction between the first layer **310** and the second layer **320**. Referring to FIG. 5A, the capillary pattern **300** may include a plurality of curved units (curved portions) **303**. The plurality of curved units **303** open toward the surface of the semiconductor chip **100**, for example, the upper surface **101**, and are convex in a direction opposite to the upper surface **101**. An amount of change in the curvature of the capillary pattern **300** according to temperature is changed by a size, spacing, number, etc. of the curved units **303**. The size, spacing, number, etc. of the curved units **303** may be determined so that the curvature of the capillary pattern **300** is in an appropriate state for cooling the semiconductor chip **100** according to temperature. Because a space between the curved units **303** and the upper surface **101** of the semiconductor chip **100** acts as a capillary channel, even if a gap between the capillary pattern **300** and the upper surface **101** of the semiconductor chip **100** is narrow, a capillary channel having a sufficient size is formed, and thus, the liquid coolant LC may be sufficiently supplied to the upper surface **101** of the semiconductor chip **100**. In addition, the curved units **303** may function as an anti-distortion unit that reduces or prevents the capillary pattern **300** from being distorted in a width direction when the curvature changes as the capillary pattern **300** is curved. To this end, the curved units **303** may extend in a width direction perpendicular to the length direction of the capillary pattern **300**.

(46) FIG. 5B shows a capillary pattern **300** according to an embodiment. In order to clarify the characteristics of the embodiment, in FIG. 5B, the capillary pattern **300** is simply shown without distinction between the first layer **310** and the second layer **320**. Referring to FIG. 5B, when sagging of the capillary pattern **300** occurs, the capillary pattern **300** may contact the upper surface **101** of the semiconductor chip **100**. When stiction occurs due to a surface contact between the capillary pattern **300** and the upper surface **101** of the semiconductor chip **100**, an abnormal behavior, such as the capillary pattern **300** not bending or being distorted, may occur even when the temperature changes. The capillary pattern **300** according to an embodiment includes an anti-adhesion unit. As an example, the anti-adhesion unit may include a plurality of protrusions **304** formed on, for example, a surface of the capillary pattern **300** facing the upper surface **101** of the semiconductor chip **100**. Therefore, even if sagging of the capillary pattern **300** occurs, because the plurality of protrusions **304** contact the upper surface **101** of the semiconductor chip **100**, a surface contact between the capillary pattern **300** and the upper surface **101** of the semiconductor chip **100** and resulting adhesion may be minimized or prevented. In addition, because the plurality of

protrusions **304** form a plurality of fine capillary channels therebetween and thus form a thin and uniform coolant film on the upper surface **101** of the semiconductor chip **100**, cooling efficiency may be improved.

(47) FIG. 5C shows a capillary pattern **300** according to an embodiment. In order to clarify the characteristics of the embodiment, the capillary pattern **300** in FIG. 5C is simply shown without distinction between the first layer **310** and the second layer **320**. Referring to FIG. 5C, as an embodiment, the anti-adhesion unit may include a surface roughness structure **305** on a surface of the capillary pattern **300**, for example, a surface facing the upper surface **101** of the semiconductor chip **100**. Accordingly, an effect similar to that of the plurality of protrusions **304** described above may be obtained.

(48) In the above, the capillary pattern **300** formed on the upper surface **101** of the semiconductor chip **100** has been described, and the above descriptions may also be equally applied to the capillary pattern **300** formed on the upper surfaces **101a** and **101b** of the semiconductor chips **100a** and **100b**.

(49) The cooling channel **200** may also be formed inside the semiconductor chip **100**. FIG. 6 is a schematic diagram of a configuration of the semiconductor device **1** according to an embodiment. The semiconductor device **1** shown in FIG. 6 is different from the semiconductor device **1** shown in FIG. 1 in that a cooling channel **200** is formed even inside the semiconductor chip **100**, that is, inside the substrate **110**. Hereinafter, differences will be mainly described below, and the descriptions given with reference to FIGS. 1 to 5 may also be applied to the embodiment shown in FIG. 6.

(50) Referring to FIG. 6, an immersed unit **130** (e.g., a groove) immersed from the upper surface **101** is provided in the semiconductor chip **100**. For example, the immersed unit **130** may be formed by partially etching the substrate **110**. The cooling channel **200** includes an immersed unit **130**. The capillary pattern **300** may be formed on the immersed unit **130**, for example, on a bottom **131** of the immersed unit **130**. According to the configuration, the surface of the semiconductor chip **100** is the bottom **131** of the immersed unit **130**. The bottom **131** of the immersed unit **130** is closer to the semiconductor integrated circuit **120** than the upper surface **101** of the semiconductor chip **100**. As a result, because a coolant may be supplied to a location close to the semiconductor integrated circuit **120** which is a heat source, the semiconductor chip **100** may be more effectively cooled. The cooling channel **200** may include an upper surface **101** of the semiconductor chip **100** and the immersed unit **130**. In this case, the surface of the semiconductor chip **100** includes the upper surface **101** and the bottom **131** of the immersed unit **130**.

(51) According to the embodiments described above, a semiconductor device employing a two-phase liquid cooling structure capable of effectively supplying a liquid coolant to a heat transfer surface may be implemented. In addition, a semiconductor device employing a two-phase liquid cooling structure capable of coping with a decrease in cooling efficiency due to vapor adsorption to a heat transfer surface may be implemented.

(52) The semiconductor device of the disclosure has been described with reference to the embodiments shown in the drawings. However, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure. Therefore, the embodiments should be considered in descriptive sense only and not for purposes of limitation. The scope of the disclosure is defined not by the detailed description of the disclosure but by the appended claims, and all differences within the scope will be construed as being included in the disclosure.

(53) It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and

details may be made therein without departing from the spirit and scope as defined by the following claims.

Claims

1. A semiconductor device comprising: a semiconductor chip comprising a semiconductor integrated circuit; a cooling channel comprising a surface of the semiconductor chip and configured to provide a passage for a coolant to cool the semiconductor chip; and a plurality of flexible capillary patterns on the surface of the semiconductor chip inside the cooling channel and configured to move the coolant by capillary action, wherein each capillary pattern of the plurality of flexible capillary patterns comprises a first portion in a length direction of the capillary pattern that contacts and is supported by the surface of the semiconductor chip, and a second portion in the length direction that is spaced apart from and unsupported by the surface of the semiconductor chip, wherein a curvature of the second portion of each capillary pattern of the plurality of flexible capillary patterns changes according to temperature, and wherein at least one capillary pattern of the plurality of flexible capillary patterns comprises a plurality of protrusions configured to prevent adhesion with the surface of the semiconductor chip.
2. The semiconductor device of claim 1, wherein each capillary pattern of the plurality of flexible capillary patterns further comprises a first layer having a first thermal expansion coefficient and a second layer having a second thermal expansion coefficient that is different from the first thermal expansion coefficient.
3. The semiconductor device of claim 1, wherein the second portion of each capillary pattern of the plurality of flexible capillary patterns has a roll shape.
4. The semiconductor device of claim 1, wherein the surface of the semiconductor chip on which the plurality of flexible capillary patterns are formed is an upper surface of the semiconductor chip.
5. The semiconductor device of claim 1, wherein each capillary pattern of the plurality of flexible capillary patterns comprises a first layer having a first thermal expansion coefficient and a second layer having a second thermal expansion coefficient that is less than the first thermal expansion coefficient, and wherein the first layer is adjacent to the surface of the semiconductor chip, and the second layer is disposed on the first layer.
6. The semiconductor device of claim 1, wherein each capillary pattern of the plurality of flexible capillary patterns comprises a plurality of curved portions that are convex in a direction opposite to the surface of the semiconductor chip.
7. The semiconductor device of claim 6, wherein each curved portion of the plurality of curved portions extends in a width direction of the capillary pattern that is perpendicular to the length direction.
8. The semiconductor device of claim 1, wherein each capillary pattern of the plurality of flexible capillary patterns comprises a plurality of protrusions configured to prevent adhesion with the surface of the semiconductor chip.
9. The semiconductor device of claim 8, wherein the plurality of protrusions are on a surface of the capillary pattern facing the surface of the semiconductor chip.
10. The semiconductor device of claim 1, wherein each capillary pattern of the plurality of flexible capillary patterns comprises a surface roughness structure on a surface of the capillary pattern facing to the surface of the semiconductor chip.
11. The semiconductor device of claim 1, wherein the cooling channel comprises an immersed unit immersed from an upper surface of the semiconductor chip, wherein the upper surface of the semiconductor chip comprises a bottom of the immersed unit, and wherein the capillary pattern is on the bottom of the immersed unit.
12. The semiconductor device of claim 1, further comprising a housing surrounding the semiconductor chip, wherein the cooling channel is in a space between the semiconductor chip and

the housing.

13. A semiconductor device comprising: a semiconductor chip comprising a semiconductor integrated circuit; a cooling channel comprising a surface of the semiconductor chip and configured to provide a passage for a coolant to cool the semiconductor chip; and a flexible capillary pattern formed on the surface of the semiconductor chip to move the coolant by capillary action, wherein at least a portion of the flexible capillary pattern has a roll shape, wherein the capillary pattern comprises a first layer having a first thermal expansion coefficient and a second layer having a second thermal expansion coefficient that different from the first thermal expansion coefficient and wherein the flexible capillary pattern comprises an anti-distortion unit configured to prevent distortion of the flexible capillary pattern in a width direction of the flexible capillary pattern.

14. The semiconductor device of claim 13, wherein a curvature of the flexible capillary pattern changes with temperature.

15. The semiconductor device of claim 13, wherein the anti-distortion unit comprises a plurality of curved portions that are convex in a direction opposite to the surface of the semiconductor chip, and extend in the width direction of the capillary pattern.

16. The semiconductor device of claim 13, further comprising a plurality of protrusions formed on a surface of the flexible capillary pattern facing the surface of the semiconductor chip.

17. The semiconductor device of claim 13, wherein the flexible capillary pattern comprises a surface roughness structure formed on a surface of the capillary pattern facing the surface of the semiconductor chip.

18. The semiconductor device of claim 13, wherein the cooling channel comprises an immersed unit immersed from an upper surface of the semiconductor chip, wherein the upper surface of the semiconductor chip comprises a bottom of the immersed unit, and wherein the capillary pattern is on the bottom of the immersed unit.

19. The semiconductor device of claim 13, further comprising a housing surrounding the semiconductor chip, wherein the cooling channel is in a space between the semiconductor chip and the housing.
